University of Chicago **Spring 2026**
Department of Economics **ECON 31730**

Problem Set 2

Version 1 (April 3, 2026)

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The problem set should be printed out and submitted in lecture. Please separately email me your code. You are encouraged to collaborate, but please write up your answers and work on computer code independently. Use any software you like, although you learn the most by avoiding canned statistics/econometrics packages and limiting the use of generative AI. If you find any errors or require clarification, please let me know right away.

Question 1

Consider the following AR(1) model with innovations of Generalized Autoregressive Conditional Heteroskedasticity (GARCH) form:

$$\begin{aligned}
y_t &= \rho y_{t-1} + u_t, \\
u_t &= \sigma_t \varepsilon_t, \\
\sigma_t^2 &= \nu + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2, \\
\varepsilon_t &\overset{i.i.d.}{\sim} N(0,1),
\end{aligned}$$

where all random variables and parameters are scalars, and $\sigma_t > 0$. Assume that $\{y_t, \sigma_t\}$ are strictly stationary processes that satisfy the above model equations for all $t \in \mathbb{Z}$.

i) Derive $E(\sigma_t^2)$ and $\text{Var}(y_t)$.

ii) Derive $E(y_t \mid y_{t-1}, y_{t-2}, \dots)$ and $\text{Var}(y_t \mid y_{t-1}, y_{t-2}, \dots)$.

iii) Derive the likelihood function for a data set $y_1, y_2, \dots, y_T$, conditional on (y_0, y_{-1}) and $\sigma_0^2 = E(\sigma_t^2)$.

Question 2

Replicate the two figures on Slides 17–18 of the slide deck "Stationary models III".

Question 3

i) By working with the spectral density, show that there does *not* exist any univariate MA(1) process $\{y_t\}$ with autocovariances satisfying $|\Gamma_y(1)| > \Gamma_y(0)/2$.

ii) The result in (i) may seem surprising at first glance, since the Wold decomposition theorem says that any covariance stationary time series has an MA representation. How do we resolve this "paradox"?

Question 4

Let $\{y_t\}$ and $\{x_t\}$ be two mean-zero, jointly covariance stationary processes, of dimension n_y and n_x, respectively. Denote the best linear predictor of y_t given all leads and lags of $\{x_t\}$ by

$$y_t^\dagger \equiv E^*(y_t \mid \{x_\tau\}_{-\infty<\tau<\infty}) = \Theta(L)x_t, \quad \Theta(z) \equiv \sum_{\ell=-\infty}^{\infty} \Theta_\ell z^\ell.$$

Assume $\Theta(z)$ is absolutely summable (in fact, square summability is enough in the following). Assume also that $s_x(\omega)$ is nonsingular for every $\omega \in [-\pi, \pi]$. As usual, the notation $s_q(\cdot)$ denotes the spectral density matrix function of any vector series $\{q_t\}$, while the cross-spectrum between vector series $\{q_t\}$ and $\{r_t\}$ is denoted $s_{qr}(\cdot)$.

i) Show that

$$\Theta(e^{-i\omega}) = s_{yx}(\omega)s_x(\omega)^{-1} \quad \text{for all } \omega \in [-\pi, \pi].$$

ii) Show that

$$s_{y^\dagger}(\omega) = s_{yx}(\omega)s_x(\omega)^{-1}s_{xy}(\omega) \quad \text{for all } \omega \in [-\pi, \pi].$$

iii) Consider the special case $n_y = n_x = 1$. Define the *coherency* $K_{yx}(\cdot)$ between $\{y_t\}$ and $\{x_t\}$ at frequency ω as $K_{yx}(\omega) \equiv s_{yx}(\omega)/\sqrt{s_y(\omega)s_x(\omega)}$ for all $\omega \in [-\pi, \pi]$. Assuming that $s_y(\omega) > 0$ for all ω, show that

$$|K_{yx}(\omega)|^2 = \frac{s_{y^\dagger}(\omega)}{s_y(\omega)} \quad \text{for all } \omega \in [-\pi, \pi].$$

Question 5

Go to the St. Louis Federal Reserve Bank's online FRED database and download the monthly Civilian Unemployment Rate series, both seasonally adjusted (FRED ID: UNRATE) and unadjusted (FRED ID: UNRATENSA). Use the sample 1948:1–2019:12.

i) For each of the two series separately, plot an estimate of the log spectral density (across a fine grid of frequencies $\omega \in [0, \pi]$) based on least-squares estimates of a univariate AR(p) model. Estimate p by BIC, allowing for a maximum of 50 lags. Plot a 95% confidence band for the log spectral density. Compute standard errors using the delta method, and assuming the AR(p) innovations u_t are i.i.d. with $E(u_t^4) < \infty$. Ignore estimation uncertainty in p for simplicity. [Hint: As in the linear regression model, the least-squares estimator $\hat{\sigma}^2$ of the innovation variance $\mathrm{Var}(u_t)$ is asymptotically independent of the estimator of the AR slope coefficients, with variance $\mathrm{Var}(\hat{\sigma}^2) \approx \frac{1}{T-p}\,\mathrm{Var}(u_t^2)$.]

ii) For each of the two series separately, plot an estimate of the log spectral density (across a fine grid of frequencies) based on a kernel spectral estimator with quadratic spectral (Epanechnikov) kernel. Use a bandwidth of 10 periodogram ordinates. Plot a 95% confidence band for the log spectral density using the asymptotic normal approximation. [Hint: Use your programming language's built-in Discrete Fourier Transform function to compute the periodogram.]

iii) Has the statistical agency done a good job of removing seasonality from the unemployment series? Please be brief.

Question 6

Consider estimation of the New Keynesian Phillips Curve (NKPC)

$$\pi_t = c + \gamma_f E_t(\pi_{t+1}) + \gamma_b \pi_{t-1} + \lambda x_t + \nu_t,$$

where the cost-push shock satisfies $E_{t-1}(\nu_t) = 0$, and c is a constant intercept. As in the "Frequentist inference" slide deck, we impose rational expectations and base estimation on the moment conditions

$$E\Big[(\pi_t - c - \gamma_f \pi_{t+1} - \gamma_b \pi_{t-1} - \lambda x_t) Z_{t-1}\Big] = 0_{r \times 1},$$

where Z_{t-1} is an r-dimensional vector of variables known at time $t-1$.

Go to the St. Louis Federal Reserve Bank's online FRED database and download the following two series: the quarterly GDP implicit price deflator, seasonally adjusted (FRED ID: GDPDEF); and the civilian unemployment rate, seasonally adjusted (FRED ID: UNRATE). Transform the unemployment rate to quarterly averages. Use the largest available overlapping data sample for the two quarterly series, but drop all data in and after 2020.

We identify π_t with the quarter-over-quarter log growth rate of the GDP deflator, and identify the economic slack variable x_t with the level of the quarter-average unemployment rate. Let the vector Z_{t-1} contain a constant, lags 1–3 of π_t, and lags 1–3 of x_t (so $r = 7$).

i) Estimate the parameters c, γ_f, γ_b, and λ by GMM with a weight matrix equal to the inverse of the sample second moment matrix of Z_t (which yields the two-stage least squares estimator).

ii) Use the result in (i) to estimate the above-mentioned parameters by efficient (two-step) GMM. To this end, you may use any estimator of the sandwich matrix that is consistent with weakly dependent data, but you should state clearly how your estimator is defined.

iii) Report standard errors for your estimates in (i) and (ii).

iv) Report the p-value of the GMM over-identification test (Hansen's J-test).